



Knowledge-aware evolutionary graph neural architecture search

Chao Wang, Jiaxuan Zhao, Lingling Li  , Licheng Jiao, Fang Liu, Xu Liu, Shuyuan Yang

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Abstract

Graph neural architecture search (GNAS) can customize high-performance graph neural network architectures for specific graph tasks or datasets. However, existing GNAS methods begin searching for architectures from a zero-knowledge state, ignoring the prior knowledge that may improve the search efficiency. The available knowledge base (e.g. NAS-Bench-Graph) contains many rich architectures and their multiple performance metrics, such as the accuracy (#Acc) and number of parameters (#Params). This study proposes exploiting such prior knowledge to accelerate the multi-objective evolutionary search on a new graph dataset, named knowledge-aware evolutionary GNAS (KEGNAS). KEGNAS employs the knowledge base to train a knowledge model and a deep multi-output Gaussian process (DMOGP) in one go, which generates and evaluates transfer architectures in only a few GPU seconds. The knowledge model first establishes a dataset-to-architecture mapping, which can quickly generate candidate transfer architectures for a new dataset. Subsequently, the DMOGP with architecture and dataset encodings is designed to predict multiple performance metrics for candidate transfer architectures on the new dataset. According to the predicted metrics, non-dominated candidate transfer architectures are selected to warm-start the multi-objective evolutionary algorithm for optimizing the #Acc and #Params on a new dataset. Empirical studies on NAS-Bench-Graph and five real-world datasets show that KEGNAS swiftly generates top-performance architectures, achieving 4.27% higher accuracy than advanced

evolutionary baselines and 11.54% higher accuracy than advanced differentiable baselines. In addition, ablation studies demonstrate that the use of prior knowledge significantly improves the search performance.

Introduction

Graph neural networks (GNN) are promising tools for learning representations of graph-structured data and are utilized in numerous real-world applications, such as robustness analysis, computer vision, and time-series prediction [1]. GNNs propagate feature information between neighboring nodes to learn deeper feature representations. Common GNNs include the graph convolutional network (GCN) [2], graph sample and aggregate (GraphSAGE) [3], graph attention network (GAT) [4], and graph isomorphism network (GIN) [5]. The main differences among these models lie in how they aggregate feature information between neighboring nodes and encode feature information [6]. Owing to their excellent performance and broad range of real-world applications, research on the general architectures of GNNs continues to grow. The abundance of graph representation learning techniques makes it challenging to determine an ideal GNN model for a new graph task manually. This process is time consuming and laborious, necessitating significant expertise [7].

Graph neural architecture search (GNAS) aims to automate the customization of the optimal GNN architecture for a given graph task, providing promising directions to address the above difficulties. GNAS typically involves exploring a large search space for possible GNN architectures, evaluating their performance on the task, and using optimization algorithms to determine the best architecture. This implies that the investigation of GNAS can be categorized into two central domains: search space and search strategy. Driven by these issues, researchers have developed numerous GNAS approaches to discover high-performance architectures that surpass the manually designed architectures for various graph tasks [8], [9], [10], [11], [12]. Unlike traditional convolutional neural networks (CNNs) that operate on structured data, the GNAS search space is defined on graphs of arbitrary sizes and shapes. Therefore, the search space of GNNs is significantly larger than that of CNNs. Moreover, because graphs typically contain numerous nodes and edges, scalable GNAS methods require further research [7].

Despite its great success, GNAS still suffers from high computational costs and poor robustness. Existing work [13], [14] illustrates that one-shot or differentiable methods may be faster, but they lack robustness. Conversely, black-box optimization techniques such as evolutionary algorithms (EAs) [9], [15], [16] and reinforcement learning (RL) [8], [17] are powerful, but require extensive evaluation. Transfer optimization has emerged as a promising area in the evolution community [18]. Utilizing the experience or knowledge

gained in previous optimization tasks to help the current optimization task provides a new perspective for achieving better optimization performance. Evolutionary transfer optimization (ETO) offers an increased optimizer’s capacity automatically with accumulated knowledge, motivating researchers to explore advanced optimizers with superior transferability [19], [20]. In recent years, a large-scale benchmark for GNAS (NAS-Bench-Graph) [21] has provided a wealth of architecture libraries and their multiple potentially conflicting performance metrics on benchmark datasets, such as the accuracy (#Acc) and number of parameters (#Params). Regrettably, the GNAS methods that are currently available cannot fully benefit from these valuable databases, which have the potential to improve the search efficiency significantly.

In this paper, we propose the knowledge-aware multi-objective evolutionary GNAS (KEGNAS) framework to reduce the effort required to solve new graph tasks from scratch by efficiently capturing available knowledge from databases with multiple performance metrics, such as NAS-Bench-Graph. A knowledge model is trained once on a prior database consisting of benchmark datasets and architectures to learn a dataset-to-architecture mapping, which stores the available prior knowledge. For a new dataset, the knowledge model can generate numerous candidate transfer architectures. Subsequently, because evaluating the performance of a candidate transfer architecture on a new dataset is computationally expensive, a deep multi-output Gaussian process (DMOGP) with dataset and architecture encoding is constructed. The DMOGP predicts multiple performance metrics (#Acc and #Params) of the candidate transfer architectures on the new dataset. The surrogate model is trained once on a prior database consisting of benchmark datasets, architectures, and the performance of architectures on benchmark datasets. Notably, the generation and evaluation of candidate transfer architectures require only a few GPU seconds because of the avoidance of training on the new dataset. Finally, non-dominated candidate transfer architectures are selected as prior knowledge to warm-start the multi-objective EA (MOEA). The warm-start MOEA optimizes multiple performance metrics to obtain Pareto optimal architectures for the new graph dataset.

We perform the KEGNAS on the NAS-Bench-Graph and five real-world graph datasets. The experimental results reveal that KEGNAS outperforms several state-of-the-art GNAS methods. Moreover, ablation studies show the effectiveness of prior knowledge utilization under different task similarities. In summary, the key contributions of this study are as follows¹:

- We provide an evolutionary transfer optimization perspective to improve the search efficiency of GNAS methods, which avoids the blindness of searching from the zero-knowledge state. The prior knowledge contained in NAS-Bench-Graph is leveraged to accelerate the search for high-performance architectures on new graph datasets.

- The novel KEGNAS framework is developed, which consists of three carefully designed components: a knowledge model, a DMOGP, and a warm-start MOEA. The knowledge model and DMOGP only require a few GPU seconds to generate and evaluate transfer architectures containing prior knowledge to assist multi-objective evolutionary searches on new datasets.
- The scalability and practicality of KEGNAS are demonstrated on NAS-Bench-Graph and real-world graph datasets. Ablation studies demonstrate that transfer architectures containing rich prior knowledge significantly improve the search performance.

The remainder of this paper is organized as follows. Section 2 provides an overview of related studies on GNAS and ETO, emphasizing the motivating factors for our work. In addition, some techniques used in KEGNAS are presented. Section 3 describes the problem formulation for KEGNAS. In Section 4, we describe the proposed KEGNAS framework in detail. In Section 5, the experimental results for the NAS-Bench-Graph and real-world graph datasets, along with ablation studies, are presented to demonstrate the scalability and utility of KEGNAS. Finally, we summarize our findings and potential directions for future research in Section 6.

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Section snippets

Preliminary

In recent years, there has been a growing interest in GNAS. Because the technology of GNNs is widely employed in different computing scenarios, searching the GNN architecture for a given graph task automatically is significant. In this section, preliminary knowledge of GNNs is first introduced. We then provide a brief overview of GNAS methods closely related to the technical approach in this study according to two aspects: the search space and search strategy. Next, the related work on ETO is ...

Problem formulation

Given a target dataset $\mathcal{D} = \{\mathcal{D}_{tra}, \mathcal{D}_{val}, \mathcal{D}_{tet}\}$, the multi-objective GNAS problem can be formulated as:
$$A^* \in \underset{\alpha \in \mathcal{A}}{\operatorname{argmin}} \quad \mathcal{L}(\alpha) \quad s.t. \quad w^*(\alpha) \in \underset{w \in \mathcal{W}}{\operatorname{argmin}} \quad \mathcal{F}(w; \alpha)$$

$$= (l_1(\alpha; w^*), \dots, l_m(\alpha; w^*))^T$$

where α and w denote the architecture and its associated weights, respectively. $\mathcal{A}$ and $\mathcal{W}$ are the architecture and weight spaces, respectively. $\mathcal{L} = (l_1, \dots, l_m)^T$ is the upper-level vector-valued function with m objectives on the validation set $\mathcal{D}_{val}$, such as #Acc and #Params. $\mathcal{F}$ is the loss function on the training set $\mathcal{D}_{tra}$ for a candidate ...

Proposed framework

Fig. 2 presents an overview of the KEGNAS framework, which consists of two phases: training and search. In the following, we introduce the knowledge model, DMOGP, construction of training data from NAS-Bench-Graph, and framework of KEGNAS in detail.

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Experimental studies

This section presents a series of experimental studies on NAS-Bench-Graph and five real-world graph datasets to validate the KEGNAS framework. In addition, the effectiveness of prior knowledge utilization in KEGNAS is further explored. All experiments are running with PyTorch (v. 1.10.1) and PyTorch Geometric² (v. 2.0.3) on a GPU 2080Ti (Memory: 12 GB, Cuda version:11.3). ...

Conclusions and future work

This study aimed to establish a new ETO paradigm, KEGNAS, which improves GNAS performance by transferring valuable prior knowledge. In particular, it is demonstrated that GNAS problems can benefit from knowledge bases such as NAS-Bench-Graph. After formalizing the problem, three components are proposed: the knowledge model, DMOGP, and the warm-start MOEA. By leveraging the prior knowledge in NAS-Bench-Graph, the knowledge model and DMOGP can quickly generate and evaluate candidate transfer ...

CRedit authorship contribution statement

Chao Wang: Writing – original draft, Software, Methodology, Conceptualization. **Jiaxuan Zhao:** Writing – review & editing, Visualization, Validation, Data curation. **Lingling Li:** Writing – review & editing, Visualization, Investigation. **Licheng Jiao:** Writing – review & editing, Resources, Funding acquisition. **Fang Liu:** Writing – review & editing, Resources. **Xu Liu:** Writing – review & editing, Supervision, Conceptualization. **Shuyuan Yang:** Writing – review & editing. ...

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. ...

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